

Stochastic Processes

Practice questions for the first exam

These questions refer to Chapters 1–3 of Yuri F. Saporito's *Stochastic Processes*, excluding Section 2.5.1 (Uniform Integrability). Justify every answer and state the hypotheses of any theorem you use.

Unless stated otherwise, chains are time-homogeneous, probability vectors are columns (so distributions evolve as $\nu^T \mathbf{P}$), $\mathbb{P}_i$ and $\mathbb{E}_i$ denote probability and expectation when $X_0 = i$, and $\inf \emptyset = +\infty$. For discrete-time processes, the natural filtration is $\mathcal{F}_n = \sigma(X_0, \dots, X_n)$ unless another filtration is specified.

Continuous-time Markov chains

Question 1. Generators and short-time behavior

Notes: Section 3.1.

Consider the family of matrices, indexed by $\theta \in \mathbb{R}$,

$$\mathbf{G}_\theta = \begin{pmatrix} -3 & 3 & 0 \\ 1 & -(1+\theta) & \theta \\ 0 & 2 & -2 \end{pmatrix}, \quad S = \{0, 1, 2\}.$$

- (a) Determine all θ for which $\mathbf{G}_\theta$ is the generator of a continuous-time Markov chain. Explain both necessity and sufficiency.
- (b) For every admissible θ , give the distribution and mean of the first holding time in each state. Write the first-order expansions of $p_{01}(h)$, $p_{11}(h)$, and $p_{12}(h)$ as $h \downarrow 0$.
- (c) Find the coefficient of h^2 in the expansion of $p_{02}(h)$. Explain why its coefficient of h is zero, even when $\theta > 0$.
- (d) When $\theta > 0$, is $p_{02}(t) > 0$ for every $t > 0$?

Question 2. Stationarity and dependence on the initial state

Notes: Sections 3.1 and 3.2.1.

- (a) Let π be a probability vector and $\mathbf{G}$ a generator on a finite state space. Prove the equivalence

$$\pi^T \mathbf{P}_t = \pi^T \text{ for every } t \geq 0 \iff \pi^T \mathbf{G} = \mathbf{0}^T.$$

Prove the forward implication using the definition of the generator and the reverse implication using the matrix exponential series.

- (b) Find all stationary distributions for the generator

$$\mathbf{G} = \begin{pmatrix} -2 & 2 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & -3 & 3 \\ 0 & 0 & 1 & -1 \end{pmatrix} \quad \text{on } S = \{0, 1, 2, 3\}.$$

- (c) For an arbitrary initial distribution ν , find $\lim_{t \rightarrow \infty} \nu^T \mathbf{P}_t$ using the two-state transition formula. Explain precisely how this limit depends on ν .

Question 3. Random refreshment and the role of standardness

Notes: Sections 3.1 and 3.4; standard semigroups from the lectures.

Let $S = \{1, \dots, d\}$ with $d \geq 2$, and let ρ be a probability vector with $\rho_j > 0$ for every j . Define $\mathbf{K} = \mathbf{1}\rho^T$, so every row of $\mathbf{K}$ equals ρ^T . Fix $c > 0$ and set $\mathbf{G} = c(\mathbf{K} - \mathbf{I})$.

- (a) Verify that $\mathbf{G}$ is a generator and that $\mathbf{K}^2 = \mathbf{K}$. Use this identity to calculate $e^{t\mathbf{G}}$ explicitly.
- (b) Interpret the chain as follows: at the times of a Poisson process of rate c , choose a state independently with distribution ρ , allowing the chosen state to equal the current state. Find the rate of the first actual departure from state i .
- (c) Find the unique stationary distribution and the limit of $\mathbf{P}_t$ as $t \rightarrow \infty$. Justify both conclusions using your explicit formula.
- (d) Define another family by $\mathbf{Q}_0 = \mathbf{I}$ and $\mathbf{Q}_t = \mathbf{K}$ for every $t > 0$. Prove that $(\mathbf{Q}_t)_{t \geq 0}$ is a stochastic semigroup but is not standard. Does its generator exist as a matrix with finite entries? Can it be the transition semigroup of a chain with right-continuous paths for every initial state? Justify your answers.

Discrete-time Markov chains

Question 4. A chain built from independent trials

Notes: Sections 1.1, 1.2, and 1.4.

Let $(\xi_n)_{n \geq 1}$ be independent Bernoulli random variables with $\mathbb{P}(\xi_n = 1) = p \in (0, 1)$. Set $X_0 = 0$ and

$$X_{n+1} = \begin{cases} X_n + 1, & \xi_{n+1} = 1, \\ 0, & \xi_{n+1} = 0. \end{cases}$$

- (a) Prove the Markov property and give the transition matrix on $\{0, 1, 2, \dots\}$.
- (b) Find the distribution of X_n , including its mass at n . Explain the interpretation of X_n in terms of consecutive successes.
- (c) Prove irreducibility and aperiodicity. Find a stationary distribution and determine whether the chain is positive recurrent.
- (d) Starting from 0, find the distribution and mean of $T_0^+ = \inf\{n \geq 1 : X_n = 0\}$. Check your answer against the stationary mass at 0, and find $\lim_{n \rightarrow \infty} \mathbb{P}_0(X_n = j)$ for each fixed j .

Question 5. When is a function of a Markov chain Markov?

Notes: Section 1.2.

Consider the deterministic cycle $a \rightarrow b \rightarrow c \rightarrow a$, with X_0 uniform on $\{a, b, c\}$, and set $Y_n = \mathbf{1}_{\{X_n = c\}}$.

- (a) Compute $\mathbb{P}(Y_2 = 1 \mid Y_1 = 0, Y_0 = 1)$ and $\mathbb{P}(Y_2 = 1 \mid Y_1 = 0, Y_0 = 0)$. Verify that both conditioning events have positive probability, and decide whether (Y_n) is Markov.
- (b) Now let X be any finite-state Markov chain with transition matrix $\mathbf{P}$, and let $B_1, \dots, B_r$ partition its state space. Suppose that, for every u, v , $\sum_{j \in B_v} p_{ij}$ has the same value for all

$i \in B_u$. Prove that the block index $Y_n = u$ when $X_n \in B_u$ is a Markov chain for every initial distribution. Give its transition matrix.

- (c) Apply this criterion to a nearest-neighbor random walk on $\{0, 1, 2, 3\}$, with arithmetic modulo 4, which moves clockwise with probability $p \in (0, 1)$ and counterclockwise with probability $1 - p$. Use the blocks $\{0, 2\}$ and $\{1, 3\}$.

Question 6. Communicating classes, periodicity, and long-run behavior

Notes: Sections 1.3 and 1.4.

On $S = \{0, 1, 2, 3, 4\}$, consider

$$\mathbf{P} = \begin{pmatrix} 1/2 & 1/4 & 0 & 1/4 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3/4 & 1/4 \\ 0 & 0 & 0 & 1/2 & 1/2 \end{pmatrix}.$$

- Find all communicating classes, identify the closed classes, and classify every state as transient or positive recurrent. Find the period of every state.
- Find all stationary distributions. For each recurrent state i , calculate $\mathbb{E}_i[T_i^+]$, where $T_i^+ = \inf\{n \geq 1 : X_n = i\}$.
- Starting from 0, find the probability of eventually entering each closed class and the expected time to leave 0.
- Determine whether $\mathbb{P}_1(X_n = j)$ converges for every j . Compute, for every j , the almost sure limit of $n^{-1} \sum_{k=0}^{n-1} \mathbf{1}_{\{X_k = j\}}$ when the chain starts from 1, and describe these limits when it starts from 0.

Question 7. Return probabilities and the number of visits

Notes: Sections 1.2.1 and 1.3.1.

Fix a state i of a countable-state Markov chain. Starting from i , let $f_i = \mathbb{P}_i(T_i^+ < \infty)$ and $V_i = \sum_{n=0}^{\infty} \mathbf{1}_{\{X_n = i\}}$, counting the initial visit.

- Use the strong Markov property at successive return times to prove $\mathbb{P}_i(V_i \geq k) = f_i^{k-1}$ for $k \geq 1$.
- If $f_i < 1$, find the distribution and expectation of V_i . If $f_i = 1$, prove $\mathbb{P}_i(V_i = \infty) = 1$.
- Prove, allowing the value $+\infty$, that $\mathbb{E}_i[V_i] = \sum_{n=0}^{\infty} (\mathbf{P}^n)_{ii}$. Deduce the recurrence criterion involving this series.
- For $j \neq i$, express $\mathbb{E}_j[V_i]$ in terms of $\mathbb{P}_j(T_i < \infty)$ and $\mathbb{E}_i[V_i]$ when $f_i < 1$, where $T_i = \inf\{n \geq 0 : X_n = i\}$.

Question 8. A reflecting birth–death chain

Notes: Sections 1.3.2 and 1.4.1–1.4.4.

Let $p \in (0, 1)$ and $q = 1 - p$. On $S = \{0, 1, \dots\}$, set $p_{0,0} = q$, $p_{0,1} = p$, and, for $i \geq 1$, $p_{i,i+1} = p$, $p_{i,i-1} = q$; all other entries are zero.

- (a) Prove irreducibility and aperiodicity. Use the birth–death recurrence criterion to decide when the chain is recurrent.
- (b) Solve the detailed balance equations and determine exactly when the resulting weights can be normalized to a probability distribution.
- (c) Classify the chain as positive recurrent, null recurrent, or transient for each value of p . Explain why failure to normalize the weights alone does not distinguish null recurrence from transience.
- (d) In the positive recurrent case, compute the mean return time to 0 and the limiting proportion of time spent at 0.

Question 9. Adding holding probabilities and sampling every few steps

Notes: Sections 1.2 and 1.4.5.

Let $\mathbf{P}$ be the transition matrix of an irreducible chain on a finite set S , let $\alpha \in (0, 1)$, and set $\mathbf{L} = \alpha \mathbf{I} + (1 - \alpha) \mathbf{P}$.

- (a) Prove that $\mathbf{L}$ is irreducible and aperiodic and has exactly the same stationary distributions as $\mathbf{P}$. Deduce the limit of $\mathbf{L}^n$ as $n \rightarrow \infty$.
- (b) Show directly, using paths and the positive diagonal of $\mathbf{L}$, that there exists n_0 such that $(\mathbf{L}^n)_{ij} > 0$ for all i, j and all $n \geq n_0$.
- (c) For an integer $m \geq 2$, show that $Y_n = X_{mn}$ is a Markov chain with transition matrix $\mathbf{P}^m$. Give an irreducible $\mathbf{P}$ for which $\mathbf{P}^m$ is reducible.
- (d) Prove that $\mathbf{L}^m$ is irreducible for every $m \geq 1$. Explain the role of the holding probability in this conclusion.

Question 10. Absorption probabilities and mean absorption times

Notes: Sections 1.2.1 and 1.3.2.

Consider the chain on $\{0, 1, 2, 3\}$ with transition matrix

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1/4 & 1/4 & 1/2 & 0 \\ 0 & 1/2 & 1/4 & 1/4 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \tau = \inf\{n \geq 0 : X_n \in \{0, 3\}\}.$$

- (a) Show that $\mathbb{P}_i(\tau > n) \leq (3/4)^n$ for $i = 1, 2$. Deduce that τ is almost surely finite and integrable.
- (b) Use first-step conditioning to find $h(i) = \mathbb{P}_i(X_\tau = 3)$ for every state i .
- (c) Use first-step conditioning to find $t(i) = \mathbb{E}_i[\tau]$. State the boundary conditions in both systems of equations.
- (d) Starting from 1, find the expected number of visits to 2 before τ . Include a visit at time 0 when writing the equations for a general initial state.

Question 11. Stationarity, reversibility, and reversing time

Notes: Sections 1.4.1 and 1.4.2.

On $S = \{0, 1, 2\}$, with arithmetic modulo 3, a chain moves from i to $i + 1$ with probability $1/2$, to $i - 1$ with probability $1/4$, and stays at i with probability $1/4$.

- (a) Find the stationary distribution π , and determine whether the chain is reversible.
- (b) Assume $X_0 \sim \pi$. Derive $\hat{p}_{ij} = \mathbb{P}(X_{n-1} = j \mid X_n = i)$ and check that $\hat{p}_{ij} = \pi_j p_{ji} / \pi_i$.
- (c) For a fixed N , prove that $(X_{N-k})_{0 \leq k \leq N}$ is Markov with transition matrix $\hat{\mathbf{P}}$. Compare the probabilities of the paths $(0, 1, 2, 0)$ and $(0, 2, 1, 0)$ under the stationary initial distribution.
- (d) Find the mean return time to 0 and the almost sure limiting frequency of transitions from 0 to 1.

Conditional expectation and discrete-time martingales

Question 12. Conditional expectation as information is revealed

Notes: Sections 2.1 and 2.2.

Let $\Omega = \{1, 2, 3, 4\}$ with equally likely outcomes, and let $Z(1) = 0$, $Z(2) = 2$, $Z(3) = 4$, $Z(4) = 10$. Let $\mathcal{F}_0 = \{\emptyset, \Omega\}$, $\mathcal{F}_1 = \sigma(\{1, 2\}, \{3, 4\})$, and $\mathcal{F}_2 = 2^\Omega$.

- (a) Compute $M_n = \mathbb{E}[Z \mid \mathcal{F}_n]$ for $n = 0, 1, 2$, and verify the defining integral identity for M_1 on every atom of $\mathcal{F}_1$.
- (b) Verify the martingale property directly. Then prove the same property for $M_n = \mathbb{E}[Z \mid \mathcal{F}_n]$ for any integrable Z and any filtration.
- (c) Compute $\mathbb{E}[Z^2 \mid \mathcal{F}_1]$ and compare it with M_1^2 . Explain the connection with conditional Jensen's inequality.
- (d) Let $\mathcal{H} = \sigma(\{1, 3\}, \{2, 4\})$. Compute $\mathbb{E}[\mathbb{E}[Z \mid \mathcal{F}_1] \mid \mathcal{H}]$ and $\mathbb{E}[Z \mid \mathcal{H}]$. Explain why the tower property does not require these to agree.

Question 13. Constructing and recognizing martingales

Notes: Section 2.2.1.

Let $(\xi_n)_{n \geq 1}$ be iid, with mean μ , variance $\sigma^2 < \infty$, and moment generating function $\varphi(\theta) = \mathbb{E}[e^{\theta \xi_1}]$ finite at a specified $\theta \in \mathbb{R}$. Set $S_0 = 0$, $S_n = \sum_{k=1}^n \xi_k$, and $\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$.

- (a) Prove that $S_n - n\mu$ and $(S_n - n\mu)^2 - n\sigma^2$ are martingales. Verify adaptedness and integrability as well as the conditional expectation identities.
- (b) Prove that $e^{\theta S_n} / \varphi(\theta)^n$ is a martingale. If $\mathbb{P}(\xi_1 = 1) = p$ and $\mathbb{P}(\xi_1 = -1) = 1 - p$, with $p \in (0, 1) \setminus \{1/2\}$, find the nonzero θ for which $e^{\theta S_n}$ itself is a martingale.
- (c) Explain why $|S_n - n\mu|$ is a submartingale. Give an example showing that a martingale squared need not be a martingale.
- (d) Let Z take values -1 and 1 with equal probability, and define $Y_0 = 0$, $Y_1 = Z$, $Y_n = -Z$ for every $n \geq 2$. Show that (Y_n) has constant expectation but is not a martingale with respect to its natural filtration.

Question 14. Predictable strategies and stopped martingales

Notes: Sections 2.2.2 and 2.3.

Let (M_n) be a martingale and let H_k be a bounded, $\mathcal{F}_{k-1}$ -measurable random variable for each $k \geq 1$. Define $W_0 = M_0$ and $W_n = M_0 + \sum_{k=1}^n H_k(M_k - M_{k-1})$.

- (a) Prove that (W_n) is a martingale. Explain precisely where measurability with respect to $\mathcal{F}_{k-1}$ is used.
- (b) For a stopping time τ , show that $H_k = \mathbf{1}_{\{\tau \geq k\}}$ satisfies the hypotheses and that the resulting process is $M_{n \wedge \tau}$.
- (c) For a symmetric simple random walk S_n with increments $\xi_k \in \{-1, 1\}$, set $H_k = \xi_k$ instead. Compute $\sum_{k=1}^n H_k(S_k - S_{k-1})$ and explain why this does not contradict part (a).

Question 15. Removing the predictable drift

Notes: Section 2.2.2, Doob's decomposition.

Let (Y_n) be an integrable adapted process. Define $A_0 = 0$ and

$$A_n = \sum_{k=0}^{n-1} (\mathbb{E}[Y_{k+1} \mid \mathcal{F}_k] - Y_k), \quad M_n = Y_n - A_n.$$

- (a) Prove that (M_n) is a martingale and that A_n is $\mathcal{F}_{n-1}$ -measurable for $n \geq 1$.
- (b) If (Y_n) is a submartingale, prove that (A_n) is increasing. Prove uniqueness of the decomposition among decompositions with $A_0 = 0$ and predictable A .
- (c) Suppose $M_0 = 0$ and (M_n) is a square-integrable martingale. Apply the construction to $Y_n = M_n^2$ and express A_n using $\mathbb{E}[(M_{k+1} - M_k)^2 \mid \mathcal{F}_k]$.
- (d) Deduce $\mathbb{E}[M_n^2] = \mathbb{E}[A_n]$ and explain why distinct martingale increments are orthogonal in L^2 .

Question 16. Harmonic functions and eigenfunction martingales

Notes: Sections 1.2, 2.1, and 2.2.

Let (X_n) be a finite-state Markov chain with transition matrix $\mathbf{P}$, and identify a function $f : S \rightarrow \mathbb{R}$ with a column vector.

- (a) If $\mathbf{P}f = \lambda f$ with $\lambda \neq 0$, prove that $\lambda^{-n}f(X_n)$ is a martingale with respect to the natural filtration.
- (b) Apply this to $\mathbf{P} = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}$ on $\{0, 1\}$ and $f(0) = 1, f(1) = -1$. Find $\mathbb{E}_i[f(X_n)]$ and $\mathbb{P}_i(X_n = 0)$.
- (c) Prove that every harmonic function ($\mathbf{P}f = f$) on a finite irreducible chain is constant. Use a state at which f attains its maximum and propagate equality along paths.
- (d) Let $B \subset S$, $\tau_B = \inf\{n \geq 0 : X_n \in B\}$, and suppose h is harmonic on $S \setminus B$. Prove that $h(X_{n \wedge \tau_B})$ is a martingale. If $\tau_B < \infty$ almost surely, show that $h(i) = \mathbb{E}_i[h(X_{\tau_B})]$.

Question 17. Stopping times and the information available at stopping

Notes: Section 2.3.

- (a) For a real-valued adapted process (X_n) , a Borel set B , and a fixed integer N , show that $\tau_B = \inf\{n \geq 0 : X_n \in B\}$ is a stopping time. Show that $L = \max\{0 \leq n \leq N : X_n \in B\}$, with $\max \emptyset = 0$, need not be a stopping time, by giving a finite example.
- (b) For a stopping time τ , prove the equivalence

$$A \in \mathcal{F}_\tau \iff A \in \mathcal{F} \text{ and } A \cap \{\tau = k\} \in \mathcal{F}_k \text{ for every } k \geq 0.$$

Here $\mathcal{F}_\tau = \{A \in \mathcal{F} : A \cap \{\tau \leq n\} \in \mathcal{F}_n \text{ for every } n\}$.

- (c) For an almost surely finite τ , define $X_\tau = 0$ on $\{\tau = \infty\}$ if necessary. Prove that X_τ is $\mathcal{F}_\tau$ -measurable. Prove also that $\mathcal{F}_\sigma \subseteq \mathcal{F}_\tau$ when $\sigma \leq \tau$ are stopping times.
- (d) Let (M_n) be a martingale and $\sigma \leq \tau \leq N$ be stopping times. By splitting an event in $\mathcal{F}_\sigma$ over $\{\sigma = k\}$, first prove $M_\sigma = \mathbb{E}[M_N | \mathcal{F}_\sigma]$. Deduce $\mathbb{E}[M_\tau | \mathcal{F}_\sigma] = M_\sigma$. Justify integrability of the stopped variables.

Question 18. Gambler's ruin by optional stopping

Notes: Sections 1.3.2, 2.2.1, and 2.3.

Let $S_n = i + \sum_{k=1}^n \xi_k$, where $1 \leq i \leq b-1$, $b \geq 2$ is an integer, and the iid increments satisfy $\mathbb{P}(\xi_k = 1) = p$, $\mathbb{P}(\xi_k = -1) = q = 1 - p$, with $p \in (0, 1)$. Set $\tau = \inf\{n \geq 0 : S_n \in \{0, b\}\}$.

- (a) Find $\varepsilon > 0$ such that $\mathbb{P}_i(\tau > kb) \leq (1 - \varepsilon)^k$ for all $k \geq 0$. Deduce $\mathbb{E}_i[\tau] < \infty$.
- (b) For $p = 1/2$, apply optional stopping first to $S_{n \wedge \tau}$ and then to $S_{n \wedge \tau}^2 - (n \wedge \tau)$ to find $\mathbb{P}_i(S_\tau = b)$ and $\mathbb{E}_i[\tau]$. Justify each passage to the limit.
- (c) For $p \neq 1/2$, use the martingale $(q/p)^{S_n}$ to find $\mathbb{P}_i(S_\tau = b)$, and then use $S_n - (p - q)n$ to find $\mathbb{E}_i[\tau]$. State which version of optional stopping applies.
- (d) Check that your hitting probabilities satisfy the first-step recurrence and its boundary conditions.

Question 19. Why almost sure finiteness is not enough

Notes: Sections 1.3.2 and 2.3.

Let (S_n) be a symmetric simple random walk on $\mathbb{Z}$ starting at 0, and set $\tau = \inf\{n \geq 0 : S_n = 1\}$.

- (a) Use recurrence to explain why $\tau < \infty$ almost surely. Compute $\mathbb{E}[S_{n \wedge \tau}]$ and $\mathbb{E}[S_\tau]$.
- (b) Show that $S_{n \wedge \tau} \rightarrow S_\tau$ almost surely but not in L^1 . Identify the invalid step in an argument claiming $\mathbb{E}[S_\tau] = \lim_n \mathbb{E}[S_{n \wedge \tau}]$.
- (c) Prove $\mathbb{E}[\tau] = \infty$ by assuming the contrary and applying the optional stopping theorem for bounded increments.
- (d) Prove that no integrable random variable can dominate $|S_{n \wedge \tau}|$ for every n .

Question 20. Maximal inequalities and convergence of a random series

Notes: Sections 2.4 and 2.5, before Section 2.5.1.

Let (ξ_k) be independent random variables with $\mathbb{P}(\xi_k = 1) = \mathbb{P}(\xi_k = -1) = 1/2$. For deterministic real numbers a_k , let $M_0 = 0$ and $M_n = \sum_{k=1}^n a_k \xi_k$.

- (a) Show that (M_n) is a square-integrable martingale and calculate $\mathbb{E}[M_n^2]$.
- (b) Use Doob's maximal inequality applied to M_k^2 to show

$$\mathbb{P}\left(\max_{0 \leq k \leq n} |M_k| \geq r\right) \leq \frac{\sum_{k=1}^n a_k^2}{r^2}, \quad r > 0.$$

Also give the bound for $\mathbb{E}[\max_{0 \leq k \leq n} |M_k|^2]$ supplied by Doob's L^2 inequality.

- (c) Assume $\sum_{k=1}^{\infty} a_k^2 < \infty$. Prove that M_n converges almost surely and in L^2 to a random variable M_{∞} . Find its mean and second moment.
- (d) For $m \geq 0$, prove the tail estimate

$$\mathbb{P}\left(\sup_{n \geq m} |M_n - M_m| > r\right) \leq \frac{\sum_{k=m+1}^{\infty} a_k^2}{r^2}.$$

Apply it to $a_k = 1/k$, giving an explicit upper bound in terms of $m \geq 1$ and r .

Question 21. Almost sure convergence and loss of expectation

Notes: Section 2.5, before Section 2.5.1.

Let (η_k) be iid with $\mathbb{P}(\eta_k = 1) = r \in (0, 1)$ and $\mathbb{P}(\eta_k = 0) = 1 - r$. Define $M_0 = 1$ and $M_n = r^{-n} \prod_{k=1}^n \eta_k$.

- (a) Prove that (M_n) is a nonnegative martingale, and calculate $\mathbb{E}[M_n]$ and $\mathbb{E}[M_n^2]$.
- (b) Determine its almost sure limit M_{∞} . Check directly whether $\mathbb{E}[|M_n - M_{\infty}|] \rightarrow 0$.
- (c) State the martingale convergence theorem under the assumption $\sup_n \mathbb{E}[|M_n|] < \infty$. Explain why your example is consistent with it and why the L^2 convergence theorem does not apply.
- (d) Determine whether there is an integrable Y such that $M_n \leq Y$ for all n , and justify your answer using dominated convergence.

Challenging and tricky questions

These problems emphasize the hypotheses behind the results and combine ideas from different sections. A numerical example alone does not prove a general claim; a counterexample must satisfy all the stated hypotheses. As in the rest of the sheet, uniform integrability is not required.

Question 22. A projection whose behavior depends on the initial law

Notes: Sections 1.2 and 1.4.

Let X be a chain on $\{a, b, c\}$ with transition matrix

$$\mathbf{P} = \begin{pmatrix} 1/8 & 1/8 & 3/4 \\ 3/8 & 3/8 & 1/4 \\ 1/2 & 1/2 & 0 \end{pmatrix}, \quad Y_n = \mathbf{1}_{\{X_n=c\}}.$$

- (a) Show that the uniform distribution is stationary for X . Show also that the probabilities of moving from a and from b into the block $\{c\}$ are different.
- (b) Suppose X_0 is uniform. Prove that, conditional on any observed history $Y_0, \dots, Y_n$ of positive probability ending in $Y_n = 0$, the hidden state X_n is uniform on $\{a, b\}$. Deduce that Y is a time-homogeneous Markov chain and find its transition matrix.
- (c) Instead start from $X_0 = a$. Compare $\mathbb{P}(Y_1 = 1 \mid Y_0 = 0)$ with $\mathbb{P}(Y_2 = 1 \mid Y_1 = 0, Y_0 = 0)$. Decide whether Y is a time-homogeneous Markov chain under this initial law. Does it still satisfy the Markov property if transition probabilities are allowed to depend on time?
- (d) Explain why a condition guaranteeing that a projection is Markov for every initial law need not be necessary for one specified initial law.

Question 23. Independent recurrent coordinates need not give recurrence

Notes: Sections 1.2, 1.3.1, and 1.4.

Let $S^{(1)}, \dots, S^{(d)}$ be independent symmetric simple random walks on $\mathbb{Z}$, all starting at 0, and put $Z_n = (S_n^{(1)}, \dots, S_n^{(d)})$. You may use $\binom{2n}{n} \sim 4^n / \sqrt{\pi n}$.

- (a) Find the one-step transition probabilities of Z . Identify the communicating class E_d containing the origin; take account of the parity of every coordinate. Prove that the restriction to E_d is irreducible and find its period.
- (b) Find $\mathbb{P}(Z_n = 0)$ for odd and even n , and determine exactly for which d the origin is recurrent.
- (c) In the recurrent cases, decide whether recurrence is positive or null. One possible approach is to combine $\mathbb{P}(Z_n = 0) \rightarrow 0$ with the ergodic theorem for the occupation frequency of a positive recurrent state.
- (d) Deduce an example of two independent irreducible recurrent chains whose product, restricted to a communicating class, is transient. The two chains need not have the same state space.

Question 24. A boundary equation with more than one bounded solution

Notes: Sections 1.2.1 and 1.3.2.

Consider the chain on $\{0, 1, 2, \dots\}$ with 0 absorbing and, for $i \geq 1$, transition probabilities $p_{i,i+1} = p$ and $p_{i,i-1} = q = 1 - p$, where $p \in (1/2, 1)$. Let $\tau_0 = \inf\{n \geq 0 : X_n = 0\}$ and $h(i) = \mathbb{P}_i(\tau_0 < \infty)$.

- (a) Find every bounded function $u : \{0, 1, \dots\} \rightarrow \mathbb{R}$ satisfying

$$u(0) = 1, \quad u(i) = p u(i+1) + q u(i-1), \quad i \geq 1.$$

Which of these functions take all their values in $[0, 1]$?

- (b) Compute $h(i)$ by first finding the probability of hitting 0 before $b > i$ and then letting $b \rightarrow \infty$. Justify the limiting event, not just the limit of the formula.
- (c) Prove that h is the smallest nonnegative solution of the displayed boundary equation, even among solutions not assumed bounded. You may stop the chain at $n \wedge \tau_0$.
- (d) Explain why boundedness and the boundary value at 0 do not identify the hitting probability uniquely. Give an additional condition at infinity that does identify it.

Question 25. An integrable stopping time with a nonintegrable payoff

Notes: Sections 2.2 and 2.3.

Let $(\xi_k)_{k \geq 1}$ be iid symmetric random signs, and let N be independent of the entire sequence, with $\mathbb{P}(N = n) = 2^{-n}$ for $n \geq 1$. Set

$$\mathcal{F}_n = \sigma(N, \xi_1, \dots, \xi_n), \quad M_0 = 0, \quad M_n = \sum_{k=1}^n 2^{k-1} \xi_k.$$

Thus the value of N is known at time 0.

- (a) Prove that M is a martingale and that $\tau = N$ is a stopping time with finite expectation.
- (b) Find the distribution of M_n and compute $\mathbb{E}[|M_n|]$ for $n \geq 1$. Use independence to determine $\mathbb{E}[|M_\tau|]$.
- (c) Determine $\mathbb{E}[(M_\tau)^+]$ and $\mathbb{E}[(M_\tau)^-]$, where $x^+ = \max(x, 0)$ and $x^- = \max(-x, 0)$. Is $\mathbb{E}[M_\tau]$ a well-defined extended real number?
- (d) For each deterministic m , optional stopping gives $\mathbb{E}[M_{\tau \wedge m}] = 0$. Explain why neither this identity nor $\mathbb{E}[\tau] < \infty$ permits an optional stopping conclusion at τ . Identify the missing increment bounds in the relevant versions of the theorem.

Question 26. Revealing the endpoint changes the martingale property

Notes: Sections 2.1 and 2.2.

Let $S_0 = 0$ and $S_n = \xi_1 + \dots + \xi_n$, where the ξ_k are iid symmetric random signs. Fix an integer $N \geq 4$ and consider

$$\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n), \quad \mathcal{G}_n = \sigma(S_N, \xi_1, \dots, \xi_n), \quad 0 \leq n \leq N.$$

- (a) Prove that S is a martingale with respect to $\mathcal{F}$ but not with respect to $\mathcal{G}$.

(b) For $n < N$, calculate $\mathbb{E}[\xi_{n+1} \mid \mathcal{G}_n]$ and $\mathbb{P}(\xi_{n+1} = 1 \mid \mathcal{G}_n)$. Justify the conditional symmetry of the remaining increments.

(c) Show that

$$R_n = \frac{S_N - S_n}{N - n}, \quad 0 \leq n < N,$$

is a $\mathcal{G}$ -martingale on this time interval. Why is it consistent with the martingale definition that R_0 is random?

(d) Under the conditional probability given $S_N = z$, for any possible endpoint z , find the transition probabilities of S_n . Decide whether this conditioned process is time-homogeneous. Treat the extreme endpoints $z = \pm N$ separately.

Question 27. Stationary marginals force a martingale to stay still

Notes: Sections 1.4, 2.1, and 2.2.

Let (M_n) be a real-valued martingale such that M_n has the same distribution as M_0 for every n .

(a) First assume $\mathbb{E}[M_0^2] < \infty$. Prove that $\mathbb{E}[(M_{n+1} - M_n)^2] = 0$ and deduce that, with probability one, $M_n = M_0$ for every n .

(b) Prove the same conclusion assuming only $\mathbb{E}[|M_0|] < \infty$. A possible route is to use the strictly convex function $\varphi(x) = \sqrt{1 + x^2}$ and its tangent-line inequality. Check integrability of

$$D_n = \varphi(M_{n+1}) - \varphi(M_n) - \varphi'(M_n)(M_{n+1} - M_n),$$

determine its expectation, and identify when it can vanish.

(c) Apply your result to a finite irreducible Markov chain started in its stationary distribution. Give a second proof that $\mathbf{P}f = f$ forces f to be constant, explaining the role of strictly positive stationary masses and paths between states.

Question 28. A sharp maximal bound and the failure of an endpoint inequality

Notes: Sections 2.3 and 2.4.

Let $b \geq 2$ be an integer, and let S be a symmetric simple random walk starting at 1. Set $\tau = \inf\{n \geq 0 : S_n \in \{0, b\}\}$, $M_n = S_{n \wedge \tau}$, and $M^* = \sup_{n \geq 0} M_n$.

(a) Show that M is a nonnegative martingale with $\mathbb{E}[M_n] = 1$ for every n .

(b) Prove that $\mathbb{P}(M^* \geq k) = 1/k$ for every integer $1 \leq k \leq b$. Justify optional stopping at the first visit to 0 or b .

(c) Compute $\mathbb{E}[M^*]$ and explain why this example makes the constant in the probability bound $\mathbb{P}(\sup_n M_n \geq k) \leq 1/k$ sharp.

(d) Show that no finite universal constant C can satisfy

$$\mathbb{E}\left[\max_{0 \leq n \leq N} M_n\right] \leq C \mathbb{E}[M_N]$$

for every nonnegative martingale and every deterministic N . Your argument must produce a finite b and a finite N for any proposed C ; a statement only about an infinite time horizon is not enough.

Question 29. Can a transition matrix be observed at time one?

Notes: Sections 3.1 and 3.2.1.

For $a, b \in (0, 1)$, consider the stochastic matrix

$$\mathbf{P} = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}.$$

We ask whether there exists a time-homogeneous continuous-time chain on these same two states, with a standard semigroup and a generator with finite entries, whose transition matrix at time 1 is $\mathbf{P}$.

- Writing a candidate generator as $\mathbf{G} = \begin{pmatrix} -u & u \\ v & -v \end{pmatrix}$, derive necessary and sufficient conditions on a, b for such a generator to exist.
- Whenever it exists, find the generator explicitly and decide whether it is unique among two-state generators. Relate your condition to the sign of $\det \mathbf{P}$.
- Give a matrix with all entries strictly positive that cannot occur in this way. Discuss separately $a + b = 1$ and $a + b > 1$.
- When $a + b = 1$, every row already equals the stationary distribution. Explain why this exact equilibration in one step is compatible with a discrete-time chain but cannot occur at a finite positive time in an irreducible two-state continuous-time chain with finite rates.

Question 30. Shortest paths determine the first nonzero small-time term

Notes: Section 3.1; matrix exponentials from the lectures.

Let $\mathbf{G}$ be a generator on a finite state space, with $\mathbf{P}_t = e^{t\mathbf{G}}$. Draw a directed edge $i \rightarrow j$ precisely when $i \neq j$ and $g_{ij} > 0$. Fix distinct states i, j joined by a directed path, and let d be the smallest number of edges in such a path.

- Prove that $(\mathbf{G}^k)_{ij} = 0$ for $0 \leq k < d$. Explain why negative diagonal entries cannot create a shorter path.
- Show that

$$(\mathbf{G}^d)_{ij} = \sum_{\substack{i_0=i, i_d=j \\ i_0 \rightarrow i_1 \rightarrow \dots \rightarrow i_d}} \prod_{\ell=0}^{d-1} g_{i_\ell i_{\ell+1}} > 0,$$

where the sum runs over directed paths of length d . Deduce the leading term of $p_{ij}(t)$ as $t \downarrow 0$.

- Apply your result to $p_{03}(t)$ for

$$\mathbf{G} = \begin{pmatrix} -5 & 2 & 3 & 0 \\ 0 & -4 & 1 & 3 \\ 0 & 0 & -2 & 2 \\ 1 & 0 & 0 & -1 \end{pmatrix} \quad \text{on } \{0, 1, 2, 3\}.$$

Identify every path contributing to the leading coefficient.

- If there is no directed path from i to j , prove that $p_{ij}(t) = 0$ for every $t \geq 0$. Explain why the existence of a path gives a positive leading coefficient despite the signs of other entries and coefficients in the matrix exponential.

Question 31. A deterministic change of clock can destroy homogeneity

Notes: Sections 3.1 and 3.2.1.

Let X be a two-state chain with generator $\mathbf{G} = \begin{pmatrix} -u & u \\ v & -v \end{pmatrix}$, where $u, v > 0$. Let $a : [0, \infty) \rightarrow [0, \infty)$ be continuously differentiable and strictly increasing, with $a(0) = 0$, and set $Y_t = X_{a(t)}$.

- (a) Prove the Markov property for Y and find its transition matrix from time s to time t , for $s \leq t$.
- (b) Prove that these transition matrices depend only on $t-s$ if and only if $a(t) = ct$ for some $c > 0$. Use the two-state transition formula to justify that equal transition matrices correspond to equal elapsed times.
- (c) Take $a(t) = t^2$. Compute

$$\lim_{h \downarrow 0} \frac{\mathbf{P}_{h^2} - \mathbf{I}}{h} \quad \text{and} \quad \lim_{h \downarrow 0} \frac{\mathbf{P}_{(t+h)^2 - t^2} - \mathbf{I}}{h}.$$

Explain why a zero derivative of the transition matrix from time 0 does not imply that Y stays in its initial state. Identify the missing semigroup property in that argument.

- (d) If X_0 has the stationary distribution of X , show that every Y_t has that same distribution. Must this make Y time-homogeneous? Justify your answer.